



Alok Ranjan

POSTDOCTORAL RESEARCHER | ROBOTICS

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Via Nino Bixio 33, Pisa (PI), Italy

ABOUT ME

I am a Postdoctoral Researcher in Robotics at the Research Center E. Piaggio, University of Pisa. I received my Ph.D. degree in Robotics from the University of Pisa in June 2025. From March to June 2024, I was a Visiting Ph.D. Researcher at the Imperial College London, working on the characterisation of a mountain-goat-hoof-inspired robotic foot for quadrupedal robots.

Research Interests. My research advances legged robotics research by addressing two parallel pillars: bio-inspired design and multi-modal locomotion. While my bio-inspired work translates advantageous biological foot features into robotic hardware for terrain stability, my multi-modal research integrates diverse strategies to navigate variable environments. Together, these areas define *what to design* to effectively tackle complex and unstructured terrains. This foundation has led to my current postdoctoral focus on the more fundamental question of *how to design*. By moving beyond intuition-based methods, I utilize reinforcement learning to optimize robot morphology, using task success as the driving objective to achieve more robust and efficient robotic solutions. Beyond my primary research, I mentor Ph.D. students in their academic development and help in technical preparation of research papers and grants.

EDUCATION

Ph.D. in Robotics – University of Pisa, Italy – (*Excellent cum laude*) [Thesis] Nov 2021 – Jun 2025

Thesis: Design of Bio-Inspired and Multi-Modal Features for Robot Locomotion in Uneven Terrains

Supervisors: Prof. Manolo Garabini, Dr. Franco Angelini *Project:* [Natural Intelligence] [Website]

- Developed *analytical design guidelines* for bio-inspired robotic feet and validated them through a mountain-goat-hoof-inspired articulated prototype.
- Extended the hoof design with a situated heuristic based design methodology to develop a more robust polyurethane *elastomer based uni-body foot for passive stick-slip stability*. Designed and fabricated custom test rigs to validate prototypes, integrating sensors and motion-capture systems to analyze the complex slip-stick behavior.
- Contributed to the development of *EM-Act, a modular series-elastic actuator* designed for dynamic robotic systems.
- Designed and validated *TraQuad I* and *II* (8 & 12 DOF tracked-legged robots), and oversaw the design of *OmniQuad* (a 12 DOF omni-wheeled-legged hybrid robot).
- Proposed an *evolutionary co-design framework* integrating morphology and control optimization via reinforcement learning.

M.Sc. in Biomedical Engineering – University of Groningen (RUG), the Netherlands [Thesis] Sep 2018 – Aug 2020

Thesis: Do muscle synergies in the arm change when learning to use a multi-articulate PR-controlled prosthetic hand?

- Muscle synergies (MSs) are defined as the proportional activation of a group of muscles involved in a task. Hierarchical modularity in the central nervous system allows movement construction, where MSs are hypothesized as motor control primitives that reduce the dimensionality of a movement.
- Calculated MSs from 8-channel EMG data of multiple hand movements using the *Non-Negative Matrix Factorization* algorithm in Matlab.
- Applied *similarity analysis* and *clustering techniques* to track changes in MS structure of hand muscles in 20 participants over a 15-session training program controlling a myoelectric prosthesis interface.
- Changes in MSs were used to inform the development of *modular user-training exercises* for more robust prosthesis control.

M.Tech. in Structures and Design – Indian Institute of Space Science and Technology (IIST), India [Thesis] Aug 2014 – Jun 2016

Thesis: Analysis and Design of a Rover Suspension for Mobility on Uneven Terrain

- Conducted dynamic simulations of a rocker-bogie suspension-based rover, *identifying critical failure modes* such as vehicle overturning and bogie-flipping during locomotion.
- Integrated a four-bar linkage between the rocker and bogie to *shift the instantaneous center of rotation*; this transitioned wheel-obstacle interaction from *overturning-prone to climb-prone behavior*.
- Optimized the suspension geometry to enable smooth stair traversal of obstacles up to 1.5× the wheel diameter (100 mm), resulting in a novel suspension that eliminated previous instabilities.
- Achieved *significant reductions in friction and torque requirements*, substantially improving traction and power efficiency on slippery or uneven terrains.

B.Tech. in Mechanical Engineering – Krishna Institute of Engineering and Technology, India [Thesis] Aug 2008 – Jun 2012

Thesis: Optimisation of a Disc Brake for an Automobile

- Addressed the critical trade-off in automobile disc brake design between rapid heat dissipation and structural integrity; while cross-drilling increases surface area for cooling, it can compromise the factor of safety due to stress concentrations.

- Performed a coupled thermal–structural FEM analysis across 1,250 parameterized configurations to identify the optimal disc geometry maximizing heat dissipation while maintaining a robust factor of safety.

WORK EXPERIENCE

Postdoctoral Researcher – University of Pisa, Italy

Jul 2025 – Present

Project–OCCAM (Italian MUR FISA23)

Dec 2025 – Present

Project–ARCADIT (PNRR)

Jul 2025 – Dec 2025

- Developing the state-estimator and whole-body control architecture for the *TraQuad II* hybrid locomotion platform.
- Synthesizing bio-inspired foot design principles to establish a standardized development pipeline for future quadrupedal robot end-effectors.
- Engineering a logical framework that integrates bio-inspiration and multi-modal design into an evolutionary morphology pipeline by implementing reinforcement learning to optimize robot morphology, replacing intuition-based heuristics with objective-driven, task-success metrics.
- Leading the technical preparation of research papers and mentoring Ph.D. candidates in experimental design and academic development.

Engineer – AllRound Technology BV, Enschede, the Netherlands

Mar 2021 – Nov 2021

- *Royal Eijkelkamp*: Managed supply chain operations for the **technology transfer** of deep submersible pump Accura II by identifying and **auditing new vendors** within the EU for raw material sourcing and specialized processing to re-establish production for the Dutch market.
- *Royal Eijkelkamp*: Developed a battery-operated mobile automated auger with vibration drilling to enable autonomous soil sampling at depths of up to 10 meters. Designed the **mobile auger and custom electrical circuits** for power management.
- *Barry Emons Aangepast Spelmateriaal BV*: Conducted physical testing and prepared risk management documentation for medical beds designed for elderly and differently abled individuals (IEC 60601 compliance).

Graduate Teaching Assistant – University of Groningen, the Netherlands

Dec 2019 – Feb 2020

Coursework: Mechatronics

M.Sc. Internship – Universitair Medisch Centrum Groningen, the Netherlands [\[Git\]](#) [\[Report\]](#)

May 2019 – Aug 2019

Title: *Using corticomuscular coherence to assess cortical contribution to walking gait in Parkinson's disease*

- At the cortical level, the primary motor cortex (M1) provides region-specific motor control, while the Supplementary Motor Area (SMA) coordinates bilateral limb movements during gait.
- In PD patients, decreased SMA activation has been associated with reduced step length and diminished arm swing. Coordinated limb movement and initiation contribute to a cyclic four-limb gait pattern, where arm swing may serve as a driving force (**Hypothesis**).
- Analyzed a comprehensive EEG and EMG dataset of walking activities to investigate the correlation between cortical and muscular activity in Control and Parkinson's Disease (PD) patient groups.
- A comparison between the pooled corticomuscular activity of a *control group with normal arm swings* and a *PD group with enhanced arm swings* revealed that **enhanced arm swings restore beta-range corticomuscular coherence to near-normal levels in PD patients**, enabling faster and more **stable gait with reduced tremors**.

Senior Project Fellow – Indian Institute of Space Science and Technology, India [\[PPT1\]](#) [\[PPT2\]](#)

Jul 2016 – Aug 2018

Title: *Investigation of Design and Control of Different Types of Rovers*

- Analyzed and enhanced a novel rover suspension inspired by NASA's Curiosity Rover's rocker-bogie design, integrating a **four-bar (Grashof)** mechanism to **reduce friction and torque requirements** during locomotion.
- **Designed detailed sub-assemblies** for drive and steering mechanisms, including wheel-gearbox-motor systems, structural links, and payload interfaces (3.5 kg payload within a 10 kg total mass limit).
- **Fabricated functional wheel assemblies** with custom-designed and machined gearboxes; implemented closed-loop motor control using Arduino and characterized motor load-current behavior.
- Evaluated rover performance on uneven terrain using MSC Adams integrated with a Simulink-based control interface.

Graduate Teaching Assistant – Indian Institute of Space Science and Technology, India

Dec 2014 – Jan 2016

Courseworks: Engineering Graphics, Manufacturing Processes Lab, Strength of Materials Lab

B. Tech. Internship – Bagga Link Motors Ltd., A Maruti Suzuki service Center, New Delhi, India

Jun 2010

Title: *Automobile overhauling and troubleshooting:*

- Gained hands-on experience in engine and gearbox overhauling, clutch replacement, wheel alignment, hydraulic break overhauling of passenger vehicles.

OTHER PROJECTS

High-Performance LED cooling with Natural Convection: FEM modeling (Team of 2), RUG [\[Report\]](#)

Nov 2019

- In high- performance LED lamps about 70% of the electrical energy is lost as heat. High temperature reduces efficiency and lifetime.
- Modeled and optimized a heat sink for a 100 Watt Mira S LED setup in COMSOL-Multiphysics to maximize the heat transfer from the LEDs to the ambient with natural convection (2n-fold symmetry design, n = number of fins (variable)).

Control and trajectory planning of a 6 DOF manipulator: AX-18A Smart Robotic Arm, RUG [Git]

Oct 2019

- Implemented forward and Inverse kinematics with DH parameters, calculated Jacobians, angular velocities of joints and end effector.
- Calculated singularities, implemented trajectory planning in *joint and Cartesian space*, function based end effector velocity control.

Op-Amp analog PID controller for a DC Motor, RUG [Images]

May 2019

- Designed and Soldered an operational amplifier based proportional–integral–derivative controller to control a DC motor.
- Modified feedback from the tachometer attached to the DC motor was used to define the error signal.

Below-Knee Adjustable Prosthesis – Concept (Team of 6), RUG [Report] [PPT]

Nov 2018 – Jan 2019

- Adjustability was provided as a tool to the user to tweak parameters so that users can feel the prosthesis their own.
- Stiffness on the ankle (non-linear design), variable zero angle plantar flexion, adjustable eversion and inversion, adjustable shaft height, and adjustable position with respect to the knee were the important parameters.

Design, fabrication, and control of a Cartesian 2D plotter (Team of 2), IIST [Video]

Nov 2015

Open-loop two-axis stepper control via Arduino; draws shapes and fonts with ≈ 0.2 mm precision.

BAJA SAE India: Design, fabrication, and racing of an All Terrain Vehicle (Team of 18/19) [Report] [PPT] Mar 2010 – Feb 2012
All-India Rank 5 (2011) and Rank 4 (2012).

- BAJA SAE(Society of Automotive Engineers) INDIA is national level competition gives a platform to engineering undergraduates to design, fabricate and race their all-terrain vehicle.
- 2012: **General Secretary**, SAE-KIET Collegiate Club
- 2012: Designed the chassis with improved driver ergonomics, robust rear-loops to support rear double wishbone suspension and power-train system, Finite element analysis, Material selection, Bill of materials.
- 2012: Upgraded existing negative-symmetric all wheel steering - turning radius ≈ 1.9 m.
- 2011: Fabrication of vehicle chassis and caliper mountings, Inventory and tooling manager.

SKILLS & CERTIFICATIONS

CAD & Dynamics	Autodesk Inventor Professional, SolidWorks, MSC Adams, OpenSim 4.0
Multiphysics	ANSYS Workbench, COMSOL Multiphysics
Computation	MATLAB, Simulink, Python, C++, ROS2
Prototyping	Circuit design, CNC coding, Machining (Lathe & Milling), MIG Welding, 3D printing, Soldering
Soft Skills	Teamwork, Collaboration, Critical thinking, Problem solving, Technical writing
Certifications	Control of Mobile Robots (Coursera), Introduction to Nonlinear Systems & Control (Prof. H. Khalil)

PUBLICATIONS

- [J1] **Ranjan, A.**, Angelini, F., Nanayakkara, T., & Garabini, M. (2024). Design Guidelines for Bioinspired Adaptive Foot for Stable Interaction With the Environment. *IEEE/ASME Trans. Mechatronics*, 29(2), 843–855. [DOI]
- [J2] Gopanunni, R. K. M., Martignetti, L., Iotti, F., **Ranjan, A.**, Angelini, F., & Garabini, M. (2024). EM-Act: A Modular Series Elastic Actuator for Dynamic Robots. *IEEE Open J. Industrial Electronics Society*, 5, 468–480. [DOI]
- [J3] **Ranjan, A.**, Iotti, F., Angelini, F., & Garabini, M. (2025). TraQuad: A Modular Tracked-Legged Multimodal Quadrupedal Robot. *IEEE/ASME Trans. Mechatronics*, 1–13. [DOI]
- [J4] Iotti, F., **Ranjan, A.**, Angelini, F., & Garabini, M. (2025). OmniQuad: A Wheeled-Legged Hybrid Robot with Omnidirectional Wheels. *Mechanism and Machine Theory*, 214, 106125. [DOI]
- [J5] Gopanunni, R. K. M., **Ranjan, A.**, Martignetti, L., Angelini, F., & Garabini, M. (2025). RM-Act: A Novel Modular Harmonic Actuator. *Actuators*, 14(10), 492. [DOI]
- [J6] Gopanunni, R. K. M., **Ranjan, A.**, Martignetti, L., Angelini, F., & Garabini, M. (2025). RM-Act 2.0: A Modular Harmonic Actuator Towards Improved Torque Density. *Actuators*, 14(11), 538. [DOI]
- [J7] **Ranjan, A.***, Kalogroulis, C.*, Hewett, J., Angelini, F., Garabini, M., & Nanayakkara, T. Designing Passive Stability in Mountain Goat-Inspired Robotic Feet Through Situated Heuristics. *npj Robotics*, 3(1), 41. [DOI]
- [J8] **Ranjan, A.***, Perera, S.*, Kalogroulis, C., Zhang, Z., Angelini, F., Garabini, M., Abad, S. A., & Nanayakkara, T. Bioinspired Hooves for Robotics: Structural, Material, and Functional Insights. *Bioinspiration & Biomimetics*. (under review)

- [J9] Tesei, L. B., **Ranjan, A.**, Angelini, F., & Garabini, M. HEM-Act: A Harmonic Gearbox for Mobile Robots. *IEEE/ASME Trans. Mechatronics*. (under review)
- [J10] **Ranjan, A.**, Iotti, F., Angelini, F., & Garabini, M. Design and State Estimation of TraQuad II: A 12-DOF Hybrid Tracked-Legged Quadruped Robot. *Robotics and Autonomous Systems*. (under review)

International Conferences

- [C1] **Ranjan, A.**, Angelini, F., Nanayakkara, T., & Garabini, M. (2024). Design Guidelines for Bioinspired Adaptive Foot for Stable Interaction With the Environment. *IEEE/ASME AIM, Boston, MA, USA*. (presented)
- [C2] **Ranjan, A.**, Iotti, F., Angelini, F., & Garabini, M. (2025). TraQuad: A Modular Tracked-Legged Multi-Modal Quadrupedal Robot. *IEEE/ASME AIM, Hangzhou, China*. (presented)

MULTIMEDIA PORTFOLIO

Video demonstrations associated with published research and experimental validations:

VE1 Design Guidelines for Bioinspired Adaptive Foot for Stable Interaction With the Environment. [\[View on IEEE Xplore\]](#)

VE2 Designing passive stability in mountain goat-inspired robotic feet through situated heuristics [\[View on npjRobotics\]](#)

VE3 EM-Act: A Modular Series Elastic Actuator for Dynamic Robots. [\[View on IEEE Xplore\]](#)

VE4 TraQuad: A Modular Tracked Legged Multi-modal Quadrupedal Robot. [\[View on IEEE Xplore\]](#)

VE5 OmniQuad: a Wheeled-Legged Hybrid Robot with Omnidirectional Wheels. [\[View Demonstration\]](#)

VE6 RM-Act: A Novel Modular Harmonic Actuator. [\[View on MDPI\]](#)

VE7 RM-Act 2.0: A Modular Harmonic Actuator Towards Improved Torque Density. [\[View on MDPI\]](#)

FELLOWSHIPS & RESEARCH GRANTS

2025–Present	Research grant, Project–OCCAM (Italian MUR FISA23), Dipartimento di Ingegneria dell’Informazione, UniPi
2024–2025	Research grant, Project–ARCADIT (PNRR), Dipartimento di Ingegneria dell’Informazione, UniPi
2021–2024	Doctoral research grant, Project–Natural Intelligence (Horizon 2020), Centro di Ricerca “E.Piaggio”, UniPi
2014–2016	Teaching assistantship fellowship, Dept. of Space (DoS), India – IIST
2008–2012	National Scholarship Portal – Central Sector Scheme of Scholarships (CSSS), MHRD, New Delhi
2008	Dr. Ambedkar National Merit Scholarship, Dr. B. R. Ambedkar Foundation, New Delhi

LANGUAGES

English	Full professional proficiency
Hindi	Native / bilingual
Italian	Elementary (A2)

INTERESTS

Street Photography, Cooking, Chess, Hiking, Cycling, Sufi classical and Sufi rock music.

REFERENCES

Prof. Manolo Garabini

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